Multilayer Matrix Inversion
10/22/2026

The equation 11 in "Phenomenological Treatment of Multilayer Growth" (*Oxidation of Metals*, **vol 31**, 1989, pp. 415-429) shows the inverse of an *nxn* matrix (equation 9) without giving a full mathematical proof. This paper provides a constructional proof for the inverse of the original equation 9 using matrix factorization.

Derivation of the Inverse of Matrix $\mathcal{M}$

$$\mathcal{M} = \begin{bmatrix} \frac{\nu_1}{\nu_1-\nu_2} & \frac{-\nu_2}{\nu_1-\nu_2} & 0 & \dots & 0 \\ \frac{-\nu_1}{\nu_1-\nu_2} & \frac{\nu_2(\nu_1-\nu_3)}{(\nu_1-\nu_2)(\nu_2-\nu_3)} & \frac{-\nu_3}{\nu_2-\nu_3} & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & \vdots & \frac{-\nu_{n-2}}{\nu_{n-2}-\nu_{n-1}} & \frac{\nu_{n-1}(\nu_{n-2}-\nu_n)}{(\nu_{n-2}-\nu_{n-1})(\nu_{n-1}-\nu_n)} & \frac{-\nu_n}{\nu_{n-1}-\nu_n} \\ 0 & \vdots & 0 & -\frac{\nu_{n-1}}{\nu_{n-1}-\nu_n} & \frac{\nu_{n-1}}{\nu_{n-1}-\nu_n} \end{bmatrix} \quad (1)$$

where $\nu_1 \neq \nu_2 \neq \nu_3 \neq \ldots \neq \nu_n$ are non-zero real numbers. Matrix $\mathcal{M}$ is the original equation 9.

For clarity, $\mathcal{M}$ is a tridiagonal matrix and its diagonal elements can also be written as:

$$\begin{aligned} \mathcal{M}_{1,1} &= \frac{\nu_1}{\nu_1 - \nu_2} \\ \mathcal{M}_{i,i} &= \frac{\nu_i}{\nu_{i-1} - \nu_i} + \frac{\nu_i}{\nu_i - \nu_{i+1}} \qquad (2 \leq i \leq n-1) \qquad (2) \\ \mathcal{M}_{n,n} &= \frac{\nu_{n-1}}{\nu_{n-1} - \nu_n} \end{aligned}$$

Such that:

$$\mathcal{M} = \begin{bmatrix} \frac{\nu_1}{\nu_1-\nu_2} & \frac{-\nu_2}{\nu_1-\nu_2} & 0 & \dots & 0 \\ \frac{-\nu_1}{\nu_1-\nu_2} & \frac{\nu_2}{\nu_1-\nu_2} + \frac{\nu_2}{\nu_2-\nu_3} & \frac{-\nu_3}{\nu_2-\nu_3} & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & \vdots & \frac{-\nu_{n-2}}{\nu_{n-2}-\nu_{n-1}} & \frac{\nu_{n-1}}{\nu_{n-2}-\nu_{n-1}} + \frac{\nu_{n-1}}{\nu_{n-1}-\nu_n} & \frac{-\nu_n}{\nu_{n-1}-\nu_n} \\ 0 & \vdots & 0 & -\frac{\nu_{n-1}}{\nu_{n-1}-\nu_n} & \frac{\nu_{n-1}}{\nu_{n-1}-\nu_n} \end{bmatrix} \quad (3)$$

Using Gaussian elimination, it can readily be shown that the rank of matrix $\mathcal{M}$ is the same as its dimension. Therefore, matrix $\mathcal{M}$ is invertible. Let $\mathcal{V}$ be a diagonal matrix of all ν's, $\mathcal{W}$ another matrix of ν, matrix $\mathcal{M}$ can be factored as follows:

$$\mathcal{M} = \mathcal{W}\mathcal{V} \qquad (4)$$

where

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$$\mathcal{W} = \begin{bmatrix} \frac{1}{\nu_1-\nu_2} & \frac{-1}{\nu_1-\nu_2} & 0 & \dots & 0 \\ \frac{-1}{\nu_1-\nu_2} & \frac{1}{\nu_1-\nu_2}+\frac{1}{\nu_2-\nu_3} & \frac{-1}{\nu_2-\nu_3} & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & \frac{-1}{\nu_{n-2}-\nu_{n-1}} & \frac{1}{\nu_{n-2}-\nu_{n-1}}+\frac{1}{\nu_{n-1}-\nu_n} & \frac{-1}{\nu_{n-1}-\nu_n} \\ 0 & \dots & 0 & \frac{-1}{\nu_{n-1}-\nu_n} & \frac{1}{\nu_{n-1}-\nu_n}+\frac{1}{\nu_n} \end{bmatrix} \tag{5}$$

and

$$\mathcal{V} = \begin{bmatrix} \nu_1 & 0 & 0 & \dots & 0 \\ 0 & \nu_2 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & \nu_{n-1} & 0 \\ 0 & \dots & 0 & 0 & \nu_n \end{bmatrix} \tag{6}$$

Because $\mathcal{W}$ is a real symmetric matrix, it is readily diagonalizable. With the help of an invertible matrix $\mathcal{Q}$, $\mathcal{W}$ can be diagonalized without the need for solving its eigenvalues and eigenvectors.

$$\mathcal{W} = \mathcal{Q}\mathcal{D}\mathcal{Q}^T \tag{7}$$

$$\mathcal{W} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ -1 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & -1 & 1 & 0 \\ 0 & \dots & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\nu_1-\nu_2} & 0 & 0 & \dots & 0 \\ 0 & \frac{1}{\nu_2-\nu_3} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \frac{1}{\nu_{n-1}-\nu_n} & 0 \\ 0 & \dots & 0 & 0 & \frac{1}{\nu_n} \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 & \dots & 0 \\ 0 & 1 & -1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 1 & -1 \\ 0 & \dots & 0 & 0 & 1 \end{bmatrix} \tag{8}$$

Left-multiplying a matrix by $\mathcal{Q}$, each column is replaced by the difference between itself and the column on the immediate left. Likewise, right-multiplying a matrix by $\mathcal{Q}^T$ subtracts each row of the matrix from the row immediately above it. Combining equations 4 and 7 yields:

$$\mathcal{M}^{-1} = (\mathcal{W}\mathcal{V})^{-1} = \left(\mathcal{Q}\mathcal{D}\mathcal{Q}^T\mathcal{V}\right)^{-1} = \mathcal{V}^{-1}\left(\mathcal{Q}^T\right)^{-1}\mathcal{D}^{-1}\mathcal{Q}^{-1} = \mathcal{V}^{-1}\left(\mathcal{Q}^{-1}\right)^T\mathcal{D}^{-1}\mathcal{Q}^{-1} \tag{9}$$

Matrix $\mathcal{Q}$ is indeed invertible (see the Appendix):

$$\mathcal{Q}^{-1} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \dots & 1 & 1 & 0 \\ 1 & \dots & 1 & 1 & 1 \end{bmatrix} \tag{10}$$

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Right-multiplying by matrix $\mathcal{Q}^{-1}$ performs a cumulative sum of all the columns on the right; similarly, left-multiplying by matrix $\left(\mathcal{Q}^{-1}\right)^T$ leads each row be replaced by the cumulative sum of the row all the way to the final row. Substituting the matrices and carrying out the multiplication from right to left, equation (9) becomes:

$$
\begin{aligned}
\mathcal{M}^{-1} &= \mathcal{V}^{-1}\left(\mathcal{Q}^{-1}\right)^T \mathcal{D}^{-1}\mathcal{Q}^{-1} \\
&= \mathcal{V}^{-1}\begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 0 & 1 & 1 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 1 & 1 \\ 0 & \dots & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} \nu_1-\nu_2 & 0 & 0 & \dots & 0 \\ 0 & \nu_2-\nu_3 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \nu_{n-1}-\nu_n & 0 \\ 0 & \dots & 0 & 0 & \nu_n \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \dots & 1 & 1 & 0 \\ 1 & \dots & 1 & 1 & 1 \end{bmatrix} \\
&= \mathcal{V}^{-1}\begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 0 & 1 & 1 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 1 & 1 \\ 0 & \dots & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} \nu_1-\nu_2 & 0 & 0 & \dots & 0 \\ \nu_2-\nu_3 & \nu_2-\nu_3 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \nu_{n-1}-\nu_n & \dots & \nu_{n-1}-\nu_n & \nu_{n-1}-\nu_n & 0 \\ \nu_n & \dots & \nu_n & \nu_n & \nu_n \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{\nu_1} & 0 & 0 & \dots & 0 \\ 0 & \frac{1}{\nu_2} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \frac{1}{\nu_{n-1}} & 0 \\ 0 & \dots & 0 & 0 & \frac{1}{\nu_n} \end{bmatrix}\begin{bmatrix} \nu_1 & \nu_2 & \nu_3 & \dots & \nu_n \\ \nu_2 & \nu_2 & \nu_3 & \dots & \nu_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \nu_{n-1} & \dots & \nu_{n-1} & \nu_{n-1} & \nu_n \\ \nu_n & \dots & \nu_n & \nu_n & \nu_n \end{bmatrix}
\end{aligned}
\tag{11}
$$

The final matrix multiplication gives the same matrix as equation 11 in the original paper:

$$
\mathcal{M}^{-1} = \begin{bmatrix} 1 & \frac{\nu_2}{\nu_1} & \frac{\nu_3}{\nu_1} & \dots & \frac{\nu_n}{\nu_1} \\ 1 & 1 & \frac{\nu_3}{\nu_2} & \dots & \frac{\nu_n}{\nu_2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \dots & \frac{\nu_n}{\nu_{n-1}} \\ 1 & 1 & 1 & \dots & 1 \end{bmatrix} \tag{12}
$$

■ (Q.E.D.)

Acknowledgment

The author thanks professor Brian Gleeson at the University of Pittsburgh for his comments.

Appendix

The proof of $\mathcal{Q}^{-1}$ is done by direct matrix multiplication. Again, left-multiplying a matrix by $\mathcal{Q}$ replaces each column by the difference between the column and the one immediately on the left.

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$$\mathcal{Q}\mathcal{Q}^{-1} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ -1 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & -1 & 1 & 0 \\ 0 & \dots & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \dots & 1 & 1 & 0 \\ 1 & \dots & 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 1 & 0 \\ 0 & \dots & 0 & 0 & 1 \end{bmatrix} = \mathcal{I} \tag{A1}$$

Similarly, right-multiplying a matrix by $\mathcal{Q}^T$ subtracts each row by the row immediately below it:

$$(\mathcal{Q}^{-1})^T\mathcal{Q}^T = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 0 & 1 & 1 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 1 & 1 \\ 0 & \dots & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 & \dots & 0 \\ 0 & 1 & -1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 1 & -1 \\ 0 & \dots & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 1 & 0 \\ 0 & \dots & 0 & 0 & 1 \end{bmatrix} = \mathcal{I} \tag{A2}$$

In fact, matrices $\mathcal{Q}^{-1}$ and $(\mathcal{Q}^{-1})^T$ are known as the lower and upper uni-triangular matrices. Verification of the derivation can be done by constructing matrices $\mathcal{M}$ and $\mathcal{M}^{-1}$of an arbitrary dimension *n.* The following script of *Maxima* offers a procedure for n=10.

```
n:10$
D:zeromatrix(n,n)$
V:zeromatrix(n,n)$
Q:zeromatrix(n,n)$

/* make D and Q matrices */
for i:1 thru n-1 do (D[i,i]:v[i]-v[i+1], Q[i+1,i]:-1)$
D[n,n]:v[n]$

/* make V and Q matrices */
for i:1 thru n do (V[i,i]:v[i], Q[i,i]:1)$

/* make various inverse and transpose matrices */
Q_1:invert(Q)$
Q_T:transpose(Q)$
Q_1_T:transpose(Q_1)$
D_1:invert(D)$
V_1:invert(V)$

/* M matrix */
M:Q.D_1.Q_T.V$

/* Inverse M matrix */
M_1:V_1.Q_1_T.D.Q_1$

/* Show the product as the identity matrix */
ratsimp(M.M_1);
```

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